

Optimization and Convergence Analysis: Comprehensive Hyperparameter Sweep Results

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Abstract

This technical report presents findings from an extensive hyperparameter optimization study conducted on our cross-modal fusion architecture for neurodegenerative disease prediction. Utilizing Weights & Biases (W&B) for systematic experiment tracking, we explored the parameter space across learning rates, batch sizes, and learning rate scheduling strategies. Our analysis reveals critical insights into the convergence behavior of the AdamW optimizer and identifies significant interactions between batch size selection and training stability. We document unexpected overfitting patterns in the MRI-only encoder branch and discuss the crucial role of geometric augmentation strategies in maintaining generalization performance.

1. Experimental Setup and Tracking Infrastructure

The hyperparameter sweep was orchestrated using Weights & Biases (W&B) sweep functionality, enabling parallel execution of 247 experimental configurations across our GPU cluster. Each run was tracked with comprehensive metric logging including per-epoch training loss, validation loss, gradient norms, learning rate trajectories, and modality-specific attention weights. W&B's hyperparameter importance analysis proved instrumental in identifying the most impactful parameters, revealing that learning rate and batch size accounted for 68% of the variance in final validation performance. Our sweep configuration employed Bayesian optimization with a Gaussian Process prior, allowing the search algorithm to intelligently explore the hyperparameter space rather than relying on exhaustive grid search. This approach resulted in a $3.2\times$ reduction in computational budget while achieving comparable coverage of high-performing regions in the parameter space.

2. Learning Rate Analysis: Exploring the Optimization Landscape

The learning rate sweep spanned four orders of magnitude, from $\alpha=1\times10^{-3}$ down to $\alpha=5\times10^{-6}$. This range was deliberately chosen to capture both aggressive optimization regimes that might enable rapid convergence and conservative approaches that prioritize stability over speed.

High Learning Rate Regime (1×10^{-3} to 1×10^{-4}): Configurations in this range exhibited rapid initial loss reduction, achieving 70% of final performance within the first 10 epochs. However, these runs consistently demonstrated training instability after epoch 25, characterized by oscillating validation metrics and divergent gradient magnitudes in the cross-attention layers. The attention weight distributions became increasingly peaked, suggesting the model was fixating on spurious correlations rather than learning robust multimodal representations.

Optimal Learning Rate Regime (5×10^{-5} to 1×10^{-5}): This range produced the most consistent results across multiple random seeds. Learning curves showed smooth monotonic improvement with minimal validation loss oscillation. The optimal single learning rate was identified as $\alpha=2.3\times10^{-5}$, which balanced convergence speed with final performance. Notably, this value aligns closely with theoretical predictions from the AdamW optimizer's adaptive moment estimation mechanics given our model's depth (18 transformer layers) and parameter count (87M parameters).

Conservative Learning Rate Regime ($<1\times10^{-5}$): Extremely low learning rates produced stable training curves but required excessive computational resources to reach convergence. Runs with $\alpha=5\times10^{-6}$ showed minimal improvement after epoch 100, suggesting the optimizer was trapped in flat regions of the loss landscape with insufficient momentum to escape.

3. Batch Size Effects on AdamW Optimizer Stability

The interaction between batch size and optimizer behavior proved more complex than anticipated. AdamW's adaptive learning rate mechanism theoretically provides robustness to batch size selection, but our empirical results reveal important practical considerations specific to multimodal architectures.

Small Batch Regime ($B = 8, 16$): Small batches introduced substantial gradient noise, which paradoxically provided a beneficial regularization effect during early training. However, this noise severely degraded the effectiveness of batch normalization layers in our MRI encoder, leading to unstable running statistics and poor generalization. The coefficient of variation in gradient magnitudes exceeded 0.8, causing the AdamW second moment estimates to become unreliable.

Optimal Batch Regime ($B = 32, 64$): These configurations achieved the best balance between gradient estimate quality and regularization effects. Batch size 48 emerged as optimal, providing sufficient samples for stable batch statistics while maintaining enough stochasticity to prevent overfitting. The gradient signal-to-noise ratio in this regime averaged 12.3, enabling consistent parameter updates without excessive variance.

Large Batch Regime ($B = 128, 256$): Large batches produced smooth training curves but consistently underperformed in final validation metrics. This phenomenon, often termed the "generalization gap," manifests because large batches cause the optimizer to converge to sharp minima that fail to generalize. Additionally, our multimodal fusion layer appeared particularly sensitive to this effect, with attention weights becoming overly deterministic and losing the diversity necessary for robust cross-modal reasoning.

4. OneCycleLR Scheduler: Accelerated Convergence with Trade-offs

The OneCycleLR learning rate schedule, which employs a cosine annealing strategy with a single cycle spanning the entire training duration, demonstrated remarkable convergence acceleration compared to our baseline constant learning rate approach. Models trained with OneCycleLR reached 95% of final performance in 38% fewer epochs on average.

The scheduler's warm-up phase (epochs 1-10) proved essential for stable initialization of the attention mechanisms. Without gradual learning rate increase, we observed collapsed attention patterns where the model attended exclusively to a single spatial region in the MRI data, effectively ignoring the multimodal fusion component.

However, the aggressive learning rate decay in the second half of training (epochs 50-100) introduced unexpected complications. While the global validation loss continued to improve, modality-specific analysis revealed significant overfitting in the MRI-only encoder branch beginning at epoch 50. The training loss for the imaging pathway dropped to near-zero, while validation loss for the same pathway began diverging. This branch-specific overfitting occurred despite the overall multimodal model maintaining reasonable generalization, suggesting that the high-dimensional imaging features were memorizing training samples while the clinical score pathway provided sufficient regularization to mask this effect in aggregate metrics.

5. Critical Role of Geometric Augmentation: Bob's Elastic Transformations

The overfitting phenomenon in the MRI branch would have manifested far more severely without the sophisticated augmentation pipeline developed by Bob Martinez, our Data Engineering lead. His recent implementation of elastic deformation augmentation for 3D neuroimaging data proved essential to our training stability.

Ablation studies conducted without elastic transformations showed validation loss divergence beginning at epoch 32—a full 18 epochs earlier than observed with augmentation enabled. The elastic deformation strategy, which applies spatially-varying displacement

fields to simulate realistic anatomical variability, effectively expanded our training distribution to better represent the natural variation in brain morphology.

Bob's implementation employs a Gaussian random field with correlation length $\sigma=15$ voxels and displacement magnitude $\alpha=8$ voxels, parameters carefully tuned to preserve anatomical plausibility while introducing sufficient variation to regularize the high-capacity convolutional encoder. Without this augmentation, the model's learned features became overly specific to the exact spatial configurations present in our training set, failing to generalize to the natural variation encountered in validation data.

6. Synthesis and Recommendations

Our hyperparameter sweep yielded several actionable insights for future training runs:

1. **Optimal Configuration:** Learning rate 2.3×10^{-5} batch size 48, OneCycleLR scheduler with early stopping at epoch 50 to prevent MRI branch overfitting
2. **Modality-Specific Monitoring:** Implement separate validation metrics for each encoder branch to detect branch-specific overfitting that may be masked in aggregate performance
3. **Augmentation Dependency:** Bob's elastic transformations are non-negotiable for training stability; future work should explore additional augmentation strategies to further improve generalization
4. **Scheduler Modification:** Consider implementing a two-phase training strategy where OneCycleLR accelerates initial convergence (epochs 1-50), followed by a constant low learning rate for fine-tuning (epochs 50-70)

These findings establish a robust foundation for subsequent architectural experiments and demonstrate the critical importance of systematic hyperparameter optimization in multimodal deep learning systems.

7. Conclusion

This comprehensive hyperparameter sweep, enabled by W&B's tracking infrastructure, has illuminated the complex optimization landscape of our cross-modal fusion architecture. The identification of optimal learning rate and batch size configurations, combined with insights into scheduler-induced overfitting patterns, provides clear direction for model refinement. The critical dependency on geometric augmentation underscores the importance of cross-functional collaboration—Bob's data engineering contributions proved as essential to model performance as architectural design decisions.